

GRADING RUBRIC

FINAL REPORT

2 _____ Cover page

- includes a conceptual figure of satellite(s)
- includes name of satellite on cover
- All group member's name
- Date and Course

3 _____ Introduction

- describe mission information
- overview of what will be in report

5 _____ Provide a name for your satellite system and describe its mission objectives (both primary and secondary)

5 _____ Describe Orbit chosen – semi-major axis or altitude, eccentricity, and inclination. Explain rationale for your choice.

10 _____ Sun-Earth system. Provide a summary of the Sun-Earth system, considering risks/hazards to satellite operations at two of the following GEO, MEO, or LEO. Quantify or characterize the different emissions from the Sun and the way they affect earth observation missions or damage spacecraft.

- Summary of Sun-Earth
 - charged particles – quantify or characterize
 - radiation – quantify or characterize
 - Earth's magnetic field
- Risks at your chosen orbital altitudes – include at least three of the following:
 - charging
 - ESD
 - solar pressure
 - Van Allen radiation belts
 - communication interference

5 _____ Space weather. Provide a summary of what constitutes space weather, current research efforts or operational capabilities to measure space weather, and why your customer would want to be informed on space weather events. Include projected impacts to a communications downlink from either the GEO, MEO, or LEO solution to ground stations on the earth.

- Definition/explanation of space weather
- Research or operational ways to monitor it
- Why space weather is important/pertinent
- Impact to communications and/or other subsystems

5 _____ Vacuum testing. Provide a summary of why vacuum testing would be advisable, considering risks/hazards to satellite operations at GEO, MEO, or LEO. Quantify or characterize the effects of a vacuum environment on mission accomplishment and potential damage to the spacecraft.

- Description of vacuum testing and its importance
- recommendation (should recommend to do vacuum testing unless perhaps cubesat mission)
- consideration of risk
- quantification or characterization

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10 _____ TCS

- Describe a thermal control system (TCD) that you recommend for your mission
- What ground testing you recommend, if any?
- Why you selected your TCS components
- Estimated equilibrium temperatures in the sun and eclipse

10 _____ Plasma

- Describe risks your spacecraft is expected to be subjected to due to plasma
- Propose appropriate mitigation actions you will use for your spacecraft
- Evaluate your proposed solution(s) impact on the spacecraft's subsystems

10 _____ Radiation

- Describe the risks your spacecraft is expected to be subjected to from the radiation environment.
- Propose appropriate mitigation actions you will use for your spacecraft
- Evaluate your proposed solution(s) impact on the spacecraft's subsystems

10 _____ ADACS design

- Estimate the effects of the four main disturbance torques for your satellite
 - drag
 - Gravity gradient
 - solar radiation pressure
 - magnetic field
- Propose mass and power requirements for your satellite's ADACS

3 _____ Conclusions

2 _____ References

- In text citations used
- Good reference list in consistent format

10 _____ Grammar

- justified
- flow
- formal paper guidelines
- few grammatical errors
- Label figures and graphs
- Table of contents

Include page links in TOC

10 _____ Simulations

Simulation of orbit

Power and thermal